

REPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTRE DE L'ENSEIGNEMENT SUPERIEUR
COMMISSION NATIONALE D'ORGANISATION
DES EXAMENS

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION
NATIONAL COMMISSION FOR THE ORGANISATION
OF THE EXAMS

NATIONAL EXAM OF HIGHER NATIONAL DIPLOMA PROPOSAL – 2025 SESSION

Specialty/Option: Software Engineering

Paper: Practice of Computer

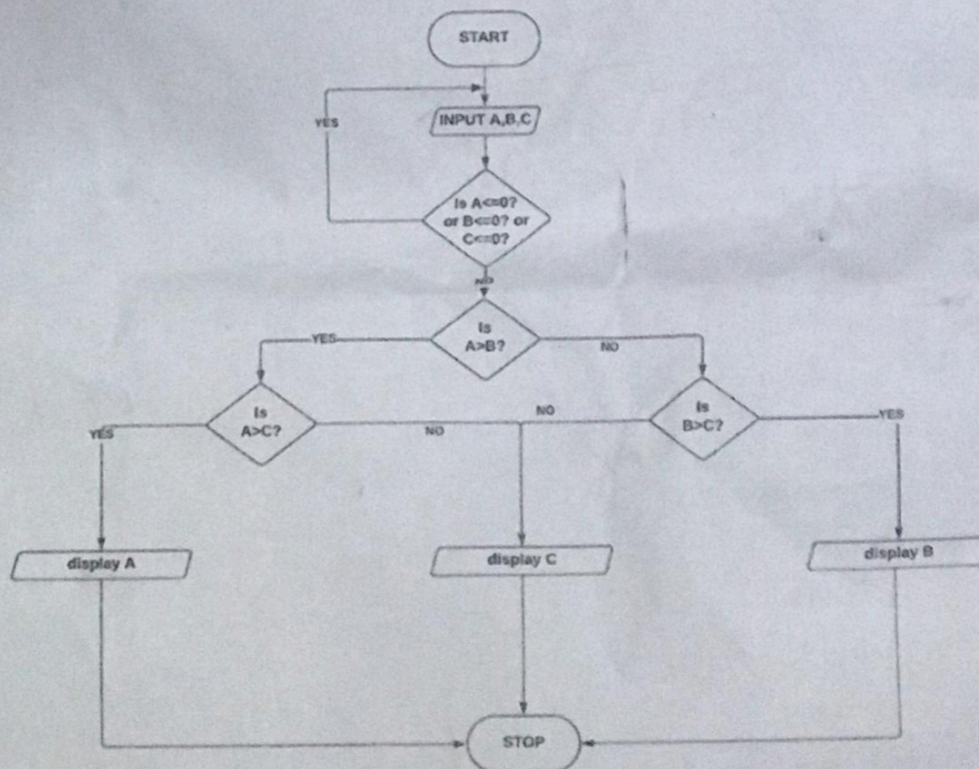
Duration: 5 hours

Credit Value: 08

SECTION A: ALGORITHM AND PROGRAMMING

/50 marks

During the last academic year, a teacher sends a student to the board to draw the following diagram:



- Q1. What is this chart representing? Define it (3 marks)
- Q2. List all the variables used and their corresponding data types (3 marks)
- Q3. Which C structure (loop or selection) can we use to convert this flowchart to code? Justify your choice (4 marks)
- Q4. List two software's which can be used to write down the corresponding C code of this flowchart (2 marks)
- Q5. Explain what the flowchart is doing (3 marks)
- Q6. After installing the appropriate software, write down the corresponding C program (20 marks)
- Q7. After installing the appropriate software, write down the equivalent java program (15 marks)

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (MYSQL) /20 marks

Given the following tables:

DNAME	DNUMBER	DMGRSSN
IT	3	237003
HR	2	237002
marketing	4	237004
administration	1	237001

DNUMBER	DLOCATION
1	Douala
1	Yaounde
3	Bafoussam
2	Douala
4	Ngaoundere
2	Maroua
3	Bamenda
4	Buea
2	Bertoua
4	Ebolowa
3	Garoua
1	Buea

Where: DNAME=Department Name

DNUMBER= Department Number
DMGRSSN= Department Manager's Social Security Number
DLOCATION= Department Location

Task 1: From the diagram provided above, draw the appropriate Entity-Relationship Diagram (4 marks)

Task 2: Using the flash provided to you, install the appropriate software (3 marks)

Task 3: select each field on the tables and provide the data type and the constraints (primary key or foreign key) (3 marks)

Task 4: Using the software installed and using SQL creates the following queries: (1 mark)

- a- create the database of the school entitled "SWE2025" (1 mark)
- b- Create the tables and named them as **table1** and **table2** including all their columns and constraints and populate with the data provided (4 marks)
- c- Display all the location present in the system (1 mark)
- d- Display all the department present at BUEA (2 marks)
- e- Count the number of department present in each town (2 marks)

Note: Each command should be copied and pasted into a Word document titled **SWE2025.docx**, which should be saved in a folder named **SWE2025** that you have created on your desktop.

/15 marks

SECTION C: WEB DESIGN

Given the following web page:

HND PRACTICAL SESSION 2025

Country	Population (in million)	
	1998	1999
YAOUNDE	85	90
	100	
	30	
BUEA	35	
	40	
	25	
BAMENDA	30	
	35	

CHECK POSITIVE OR NEGATIVE:

Input a number:

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1. Describe each element required to write and run a web page. (5 marks)
2. Differentiate between a dynamic web page and a static web page (2 marks)
3. Using appropriate software and HTML, JavaScript, or PHP, write the equivalent code for the web page described above. (8 marks)

/15marks

SECTION D: NETWORKING

As the **network administrator** for a mid-sized company, your responsibility is to design and implement a network that divides the IP address block **192.168.50.0/24** into **two subnets** using a **/25 subnet mask**. This network will serve two different departments within the organization.

You are provided with the following hardware and tools:

- 1 Router
- 2 Switches
- 6 Computers (3 per department)
- 300 meters of Ethernet cable
- Crimping tool
- RJ45 connectors

Tasks:

- 1- Describe each of the listed hardware tools and explain their specific role in the network design and implementation process (3 marks)
- 2- After installing the corresponding software found in the flash provided, design the network After installing the corresponding network simulation or configuration software provided on the USB flash drive , design the logical network topology (5 marks)
- 3- Practically implement the network design developed in question 2 (7 marks)